G H Raisoni International Skill Tech University, Pune

(Established by Maharashtra Act No. IX of 2024 and under section 2(f) of the UGC Act 1956)

FY B. TECH

ESE SUMMER 2026 (2025 Pattern)

Programming For Problem Solving (25UESL1102)

[Time: - 2.5 Hours] **[Max. Marks 60]**

Instructions to the candidates:

1) ***All questions compulsory.***
2) ***Figures to the right indicate full marks.***
3) ***Assume suitable data, if necessary.***

Q. No.	*Sub Question*	*Questions*	*Marks*	*CO*	*BL*
1	*a)*	Write a short note on Problem solving techniques.	[6]	CO2	L2
	b)	Describe Pseudo Code with an example.	[6]	CO2	L4
2	*a)*	Describe the structure of a C program.	[5]	CO4	L5
	b)	What are Keywords in C. Explain at least 5 of its types.	[3]	CO5	L4
	c)	Write a program to calculate area of a Square with a flowchart.	[4]	CO2	L6
3	*a)*	Explain Characteristics, advantages and disadvantages of control statements in C.	[5]	CO4	L5
	b)	What are the different variants of the if statements in C.	[3]	CO5	L4
	c)	Calculate if the given number is even or odd using if-else in C.	[4]	CO2	L6
4		**Solve any 2 out of the following 3 in Question number 4.**			
	a)	Elaborate 3 main aspects of Functions in C, Function Declaration, Function Definition and Function Call with Syntax	[6]	CO4	L5
	b)	Differentiate between User Defined and Library Functions with Example.	[6]	CO5	L4
	c)	Write a program to find the sum of 2 numbers using User Defined Function.	[6]	CO2	L6
5		**Solve any 2 out of the following 3 in Question number 4.**			
	a)	Define Pointers in C. Explain its Declaration and Initialization with an example.	[6]	CO4	L5
	b)	Write a C program to Swap two Numbers using Pointers.	[6]	CO5	L4
	c)	Write a program to find a program if the given number is an Armstrong number or not.	[6]	CO2	L6